

Guide to stream

stream is a Matlab function that runs the sequential analysis of tabular data. There are two types of sequence.

1. There is a sequence of analysis tasks, such as '**select plot set**'. This sequence selects a subset of the data, plots it and formats the plot.
2. For each task, the data table (see below) is split into groups which are analysed in turn. A group is defined by a set of variables in the data table, such as {'location', 'contrast'}. In this example a group is a set of rows in the table representing a single location and contrast.

Data are contained in two variables **d** and **m**, which stand for data table and metadata, respectively. A column in **d** represents a variable such as location, and a row in **d** is a single observation. The metadata variable **m** indicates how the analysis is to be performed. For example, **m.select.location = [2, 3]** indicates that the **select** task is to keep only those data table rows for which the location is [2, 3]. Task *task* is typically performed with a function call of the form **[d, m] = doTask(d, m)**, which transforms the input data table on the right into the table on the left under the instructions of the metadata on the right.

stream uses several variables:

- **d** is the data table
- **m** is the metadata
- **m.group** is the number of the group currently being analysed
- **m.groups** is the number of groups for the current task
- **m.tasks** is the sequence of tasks to be performed

- `m.task.fun` is the function used to execute *task*
- `m.task.group` is a list of the table variables that are constant within each group analysed by *task*
- `m.task.arg` is a value used by *task*
- `m.x`, `m.y`, and `m.z` are the independent and dependent variables in the analysis
- `m.handle.axes`, `m.model` are internal variables

Limited help on stream can be obtained with `help stream`. Some task functions, such as `doSelect`, include a description of their inputs and outputs.